

Home Network Configuration

Circle C Ranch residence - gigabit fiber - segmented VLANs

WAN and Core

Item	Detail
Internet	Gigabit symmetric fiber
Gateway	UniFi (10.0.20.1)
Core switch	24-port PoE (10.0.20.2)
Office uplink	Dedicated gigabit ethernet run to home office

VLAN Segmentation

VLAN	Subnet	Purpose
10 - Trusted	10.0.10.0/24	Work laptop, family devices
20 - Infra	10.0.20.0/24	Home lab, HA cluster, NAS
30 - IoT	10.0.30.0/24	60+ smart home devices, isolated
40 - Cameras	10.0.40.0/24	Ring cameras, no internet egress
50 - Guest	10.0.50.0/24	Visitor wifi, rate limited

Notes

IoT VLAN blocks lateral traffic to Trusted and Infra except through explicit Home Assistant firewall rules. Camera VLAN has no outbound internet. Powerwall and solar inverter sit on Infra for local API polling into Home Assistant.

Static Address Reservations

Infra VLAN 20 holds the home lab and smart home control plane. Addresses are reserved on the gateway so the Home Assistant cluster and storage stay stable across reboots.

Host	Address	Role
Raspberry Pi 5 - node A	10.0.20.11	Home Assistant primary
Raspberry Pi 5 - node B	10.0.20.12	HA worker, Zigbee2MQTT
Raspberry Pi 5 - node C	10.0.20.13	HA worker, failover and backups
Custom Linux desktop	10.0.20.20	Home lab host, Docker sandbox
NAS (4-bay)	10.0.20.30	Config and media backups
UniFi gateway	10.0.20.1	Router and firewall
PoE switch (24-port)	10.0.20.2	Cameras and access points

Firewall Policy Summary

Source	Destination	Action
IoT (30)	Trusted (10)	Block
IoT (30)	Infra (20)	Block except HA control port
Cameras (40)	Any internet	Block (no egress)
Guest (50)	Internal VLANs	Block, internet only
Trusted (10)	Infra (20)	Allow, management access

Wireless

Three broadcast networks map to VLANs: a trusted family SSID on VLAN 10, an isolated device SSID on VLAN 30 for smart home gear, and a rate limited guest SSID on VLAN 50 with client isolation enabled. Access points run on PoE from the core switch.

Backup and Recovery

The NAS pulls nightly snapshots of each Home Assistant node and the Linux host, with a weekly offsite copy to encrypted cloud storage. Gateway and switch configuration is exported on every change and versioned in the home lab repo.

Asset	Method	Frequency	Retention
HA node configs	Snapshot to NAS	Nightly	30 days
NAS datasets	ZFS snapshot	Nightly	14 days
Offsite copy	Encrypted cloud sync	Weekly	90 days
Network config	Config export to repo	On change	Full history

Change Log

Date	Change
2025-11-04	Split camera traffic onto isolated VLAN 40, no egress
2025-12-06	Added Powerwall gateway to Infra for local API polling
2026-01-18	Reserved node C for HA failover and backup role
2026-02-02	Tightened guest SSID rate limits and client isolation

Maintenance Notes

Firmware on the gateway and switch is reviewed monthly and applied after a NAS snapshot. The camera VLAN is verified quarterly to confirm no outbound route exists. Next scheduled review is October 2026.

Maintained by Jennifer Watson. Home lab documentation, Circle C Ranch.